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Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I will start by loading the dataset and performing an initial exploration to understand its structure and contents. Then, you can specify a specific analytical question for me to address.

Let's begin by inspecting the dataset.

The dataset contains 231,636 records and 19 columns related to municipal contracts. Key attributes include:

- **StateCode, MunicipalityCode:** Identifiers for locations.
- **MonthIssued:** Month of contract issuance.
- **ProgramCode, SubProgramCode, ResourceSourceCode:** Categorical fields representing funding sources and programs.
- **EconomicActivity:** Indicates the type of economic activity.
- **FundingValue, InvestmentValue, CommercializationValue, IndustrializationValue:** Monetary amounts associated with different contract types.
- **AllocatedAreaInvestment, AllocatedAreaFunding:** Allocated area measurements.
- **ContractTotalValue:** The total monetary value of the contract.

Now, what specific question would you like to analyze on this dataset?

What is the probability that the value allocated to industrialization will be greater than the value involved in commercialization, based on the contract's state? Define three approaches to address this issue and select the best one.

To determine the probability that the **Industrialization Value** is greater than the **Commercialization Value** for each state, we will define three approaches:

Approach 1: Empirical Probability Calculation

- For each **StateCode**, count the number of contracts where **IndustrializationValue** > **CommercializationValue**.

Approach 2: Statistical Hypothesis Testing

- Conduct a **paired sample test** (e.g., Wilcoxon signed-rank test) to see if **IndustrializationValue** is significantly higher than **CommercializationValue** for each state.
- If significant, estimate the probability from the observed differences.

Approach 3: Predictive Modeling with Logistic Regression

- Define a binary variable: 1 if **IndustrializationValue** > **CommercializationValue**, 0 otherwise.
- Train a **logistic regression model** using **StateCode** and other relevant features to predict this probability.

Choosing the Best Approach

- **Approach 2 (Statistical Testing)** is useful if we want to test significance rather than just estimate probability.
- **Approach 3 (Predictive Modeling)** can be useful if we want to generalize to unseen data but may not be necessary for a descriptive question.

Since we are interested in **descriptive probability per state**, the **Empirical Probability Calculation** (Approach 1) is the most appropriate. Now, I will compute and display the results.

Here are the top 10 states with the highest probability that the **Industrialization Value** is greater than the **Commercialization Value** in contracts:

StateCode	Probability ($P(\text{Industrialization} > \text{Commercialization})$)
7	5.15%
3	0.54%
19	0.39%
24	0.38%
1	0.38%
25	0.34%
27	0.32%
13	0.30%
14	0.23%
8	0.21%

The probability is generally low across all states, with **StateCode 7** having the highest probability at ~5.15%.
Would you like a visualization of these probabilities? [~]

Message Data Analyst

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ChatGPT can make mistakes. Check important info.